
Math 12

Intermediate Algebra

Math 12 Course at a Glance

Intermediate Algebra

Review Content

REVIEW

M12-REV-001	I can evaluate numerical expressions using the order of operations.
M12-REV-002	I can perform operations with signed integers and fractions.
M12-REV-003	I can simplify an algebraic expression by combining like terms.
M12-REV-004	I can use the distributive property to rewrite an expression.
M12-REV-005	I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED

M12-EQN-001	I can solve a linear equation and present equivalent steps clearly.
M12-EQN-002	I can solve a linear equation containing fractions.
M12-EQN-003	I can rearrange a formula to isolate a specified variable.
M12-EQN-004	I can translate a verbal relationship into a linear equation.
M12-EQN-005	I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED

M12-INE-001	I can solve and graph a one-variable linear inequality.
M12-INE-002	I can solve a compound inequality involving AND.
M12-INE-003	I can solve a compound inequality involving OR.
M12-INE-004	I can express an inequality solution using interval notation.
M12-INE-005	I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED

M12-LIN-001	I can determine and interpret the slope of a line.
M12-LIN-002	I can graph a line from an equation in a common form.
M12-LIN-003	I can write an equation of a line from given information.
M12-LIN-004	I can determine whether two lines are parallel, perpendicular, or neither.
M12-LIN-005	I can evaluate a function using function notation.
M12-LIN-006	I can determine whether a relation is a function using the vertical line test or input-output data.
M12-LIN-008	I can determine domain and range from a graph or table.
M12-LIN-009	I can interpret domain and range in context.

Math 12 Course at a Glance

Intermediate Algebra

Systems of Linear Equations

REQUIRED

M12-SYS-001	I can interpret the solution of a linear system as an intersection point.
M12-SYS-002	I can solve a two-variable linear system by graphing.
M12-SYS-003	I can solve a two-variable linear system by substitution.
M12-SYS-004	I can solve a two-variable linear system by elimination.
M12-SYS-005	I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED

M12-POL-001	I can simplify expressions using integer exponent rules.
M12-POL-002	I can add and subtract polynomials.
M12-POL-003	I can multiply polynomials, including binomials.
M12-POL-004	I can divide a polynomial by a monomial.
M12-POL-005	I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSION

M12-EXT-001	I can solve and graph an absolute-value inequality.
M12-EXT-002	I can graph a system of linear inequalities and identify its solution region.
M12-EXT-003	I can factor a quadratic trinomial.
M12-EXT-004	I can identify a relationship as discrete or continuous.

Math 12 In-Depth Curriculum

Intermediate Algebra

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

M12-REV-001	I can evaluate numerical expressions using the order of operations.
SUPPORTING SKILLS	
R	Apply the order of operations to a numerical expression. (inherited from middle school) SK-NUM-ORDER-OPERATIONS · Reinforce

M12-REV-002	I can perform operations with signed integers and fractions.
SUPPORTING SKILLS	
R	Add, subtract, multiply, and divide fractions. (inherited from middle school) SK-NUM-FRACTION-OPERATE · Reinforce
R	Add, subtract, multiply, and divide signed integers. (inherited from middle school) SK-NUM-INTEGER-OPERATE · Reinforce

M12-REV-003	I can simplify an algebraic expression by combining like terms.
SUPPORTING SKILLS	
R	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Reinforce
R	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Reinforce

M12-REV-004	I can use the distributive property to rewrite an expression.
SUPPORTING SKILLS	
R	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Reinforce

M12-REV-005	I can determine the prime factorization of a whole number.
SUPPORTING SKILLS	
R	Express a whole number as a product of prime factors. (inherited from middle school) SK-NUM-PRIME-FACTOR · Reinforce

Linear Equations and Applications

REQUIRED

M12-EQN-001

I can solve a linear equation and present equivalent steps clearly.

SUPPORTING SKILLS

I	Check a proposed solution in the original equation. SK-ALG-CHECK-SOLUTION · Introduce
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
I	Use inverse operations to maintain an equivalent equation. SK-ALG-INVERSE-OPERATIONS · Introduce
A	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Apply

M12-EQN-002

I can solve a linear equation containing fractions.

SUPPORTING SKILLS

A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
I	Clear fractions from an equation using a common denominator. SK-ALG-CLEAR-FRACTIONS · Introduce
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Add, subtract, multiply, and divide fractions. (inherited from middle school) SK-NUM-FRACTION-OPERATE · Apply

M12-EQN-003

I can rearrange a formula to isolate a specified variable.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Isolate a specified variable in an equation or formula. SK-ALG-VARIABLE-ISOLATE · Introduce

M12-EQN-004

I can translate a verbal relationship into a linear equation.

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Translate quantities and relationships into an equation. SK-ALG-TRANSLATE-EQUATION · Introduce

M12-EQN-005

I can solve and interpret a number, mixture, or motion problem using a linear equation.

SUPPORTING SKILLS

A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Deepen

Math 12 In-Depth Curriculum

Intermediate Algebra

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Inequalities and Absolute-Value Equations

REQUIRED

M12-INE-01	I can solve and graph a one-variable linear inequality.
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Represent an inequality solution on a number line. SK-INE-NUMBER-LINE · Introduce
I	Reverse an inequality when multiplying or dividing by a negative value. SK-INE-REVERSE-SIGN · Introduce

M12-INE-02	I can solve a compound inequality involving AND.
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Interpret AND as the intersection of two inequality solution sets. SK-INE-COMPOUND-AND · Introduce
A	Represent an inequality solution on a number line. (first introduced in Math 12) SK-INE-NUMBER-LINE · Apply
A	Reverse an inequality when multiplying or dividing by a negative value. (first introduced in Math 12) SK-INE-REVERSE-SIGN · Apply

M12-INE-03	I can solve a compound inequality involving OR.
SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Interpret OR as the union of two inequality solution sets. SK-INE-COMPOUND-OR · Introduce
A	Represent an inequality solution on a number line. (first introduced in Math 12) SK-INE-NUMBER-LINE · Apply
A	Reverse an inequality when multiplying or dividing by a negative value. (first introduced in Math 12) SK-INE-REVERSE-SIGN · Apply

M12-INE-04	I can express an inequality solution using interval notation.
SUPPORTING SKILLS	
A	Represent an inequality solution on a number line. (first introduced in Math 12) SK-INE-NUMBER-LINE · Apply
I	Translate between inequality and interval notation. SK-NOT-INTERVAL · Introduce

M12-INE-05	I can solve an absolute-value equation and check its solutions.
SUPPORTING SKILLS	
I	Rewrite an absolute-value equation as two cases. SK-ABS-SPLIT-EQUATION · Introduce
A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply

Math 12 In-Depth Curriculum

Intermediate Algebra

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Lines and Introductory Functions

REQUIRED

M12-LIN-001	I can determine and interpret the slope of a line.
SUPPORTING SKILLS	
I	Determine slope from a graph. SK-LIN-SLOPE-GRAPH · Introduce
I	Calculate slope from two points. SK-LIN-SLOPE-POINTS · Introduce
I	Attach and interpret units for rates, inputs, and outputs. SK-REP-UNITS · Introduce

M12-LIN-002	I can graph a line from an equation in a common form.
SUPPORTING SKILLS	
I	Convert among common forms of a linear equation. SK-LIN-FORM-CONVERT · Introduce
I	Use slope to generate additional points on a line. SK-LIN-GRAPH-SLOPE · Introduce
I	Plot an intercept accurately on a coordinate plane. SK-LIN-PLOT-INTERCEPT · Introduce

M12-LIN-003	I can write an equation of a line from given information.
SUPPORTING SKILLS	
A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Deepen
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply

M12-LIN-004	I can determine whether two lines are parallel, perpendicular, or neither.
SUPPORTING SKILLS	
I	Recognize that distinct parallel lines have equal slopes. SK-LIN-PARALLEL-SLOPE · Introduce
I	Determine the negative reciprocal of a nonzero slope. SK-LIN-PERPENDICULAR-SLOPE · Introduce

M12-LIN-005	I can evaluate a function using function notation.
SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Identify input and output values in a formula, graph, table, or context. SK-FUNC-INPUT-OUTPUT · Introduce
I	Identify the input and output represented by function notation. SK-FUNC-NOTATION-READ · Introduce

M12-LIN-006	I can determine whether a relation is a function using the vertical line test or input-output data.
SUPPORTING SKILLS	
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
I	Apply the vertical line test to a graph. SK-FUNC-VERTICAL-LINE · Introduce

M12-LIN-008	I can determine domain and range from a graph or table.
SUPPORTING SKILLS	
I	Read domain and range from a graph. SK-FUNC-DOMAIN-RANGE-GRAPH · Introduce
I	Read domain and range from a table. SK-FUNC-DOMAIN-RANGE-TABLE · Introduce

Math 12 In-Depth Curriculum

Intermediate Algebra

SKILL PROGRESSION

I

Introduce

R

Reinforce

D

Deepen

A

Apply

M12-LIN-009	I can interpret domain and range in context.
SUPPORTING SKILLS	
I	Restrict domain to values meaningful in a context. SK-FUNC-CONTEXT-DOMAIN · Introduce
I	Interpret the range of a function in a context. SK-FUNC-CONTEXT-RANGE · Introduce
I	Restrict a model to meaningful contextual inputs. SK-MODEL-VALID-DOMAIN · Introduce

Math 12 In-Depth Curriculum

Intermediate Algebra

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Systems of Linear Equations

REQUIRED

M12-SYS-001

I can interpret the solution of a linear system as an intersection point.

SUPPORTING SKILLS

I	Locate and interpret the intersection of two lines. SK-SYS-GRAPH-INTERSECTION · Introduce
---	--

M12-SYS-002

I can solve a two-variable linear system by graphing.

SUPPORTING SKILLS

A	Use slope to generate additional points on a line. (first introduced in Math 12) SK-LIN-GRAPH-SLOPE · Apply
A	Plot an intercept accurately on a coordinate plane. (first introduced in Math 12) SK-LIN-PLOT-INTERCEPT · Apply
I	Check an ordered pair in both equations of a system. SK-SYS-CHECK · Introduce
D	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Deepen

M12-SYS-003

I can solve a two-variable linear system by substitution.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
I	Substitute an equivalent expression for a variable. SK-SYS-SUBSTITUTE · Introduce

M12-SYS-004

I can solve a two-variable linear system by elimination.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
I	Add equations to eliminate a variable. SK-SYS-ELIMINATE · Introduce

M12-SYS-005

I can create and solve a two-variable linear system for an application.

SUPPORTING SKILLS

A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
A	Add equations to eliminate a variable. (first introduced in Math 12) SK-SYS-ELIMINATE · Apply
I	Define two variables and write two equations for a context. SK-SYS-MODEL · Introduce
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

Math 12 In-Depth Curriculum

Intermediate Algebra

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Polynomial Operations and Basic Factoring

REQUIRED

M12-POL-001	I can simplify expressions using integer exponent rules.
SUPPORTING SKILLS	
I	Apply the power rule for exponents. SK-EXP-POWER · Introduce
I	Apply the product rule for exponents. SK-EXP-PRODUCT · Introduce
I	Apply the quotient rule for exponents. SK-EXP-QUOTIENT · Introduce

M12-POL-002	I can add and subtract polynomials.
SUPPORTING SKILLS	
A	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Apply
I	Identify like polynomial terms by variable part and exponent. SK-POLY-LIKE-TERMS · Introduce
I	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. SK-POLY-VOCABULARY · Introduce

M12-POL-003	I can multiply polynomials, including binomials.
SUPPORTING SKILLS	
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
I	Multiply two binomials using distribution. SK-POLY-BINOMIAL-MULTIPLY · Introduce
I	Multiply a monomial by a polynomial. SK-POLY-DISTRIBUTE · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M12-POL-004	I can divide a polynomial by a monomial.
SUPPORTING SKILLS	
A	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Apply
I	Divide each term of a polynomial by a monomial. SK-POLY-DIVIDE-MONOMIAL · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M12-POL-005	I can factor a greatest common factor from a polynomial.
SUPPORTING SKILLS	
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
I	Factor the greatest common monomial factor from a polynomial. SK-FACTOR-GCF · Introduce
I	Determine the numerical and variable greatest common factor of polynomial terms. SK-POLY-GCF · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

Extension Content

EXTENSION

M12-EXT-001

I can solve and graph an absolute-value inequality.

SUPPORTING SKILLS

I	Rewrite an absolute-value inequality as a compound inequality. SK-ABS-SPLIT-INEQUALITY · Introduce
A	Interpret AND as the intersection of two inequality solution sets. (first introduced in Math 12) SK-INE-COMPOUND-AND · Apply
A	Interpret OR as the union of two inequality solution sets. (first introduced in Math 12) SK-INE-COMPOUND-OR · Apply
A	Represent an inequality solution on a number line. (first introduced in Math 12) SK-INE-NUMBER-LINE · Apply

M12-EXT-002

I can graph a system of linear inequalities and identify its solution region.

SUPPORTING SKILLS

I	Graph the boundary line of a linear inequality with the correct line type. SK-INE-GRAPH-BOUNDARY · Introduce
I	Identify the overlapping solution region of multiple linear inequalities. SK-INE-SYSTEM-REGION · Introduce
A	Use slope to generate additional points on a line. (first introduced in Math 12) SK-LIN-GRAPH-SLOPE · Apply

M12-EXT-003

I can factor a quadratic trinomial.

SUPPORTING SKILLS

I	Verify a factorization by multiplication. SK-FACTOR-CHECK · Introduce
I	Factor a quadratic trinomial with a nonunit leading coefficient. SK-FACTOR-TRINOMIAL-GENERAL · Introduce
I	Factor a monic quadratic trinomial. SK-FACTOR-TRINOMIAL-MONIC · Introduce

M12-EXT-004

I can identify a relationship as discrete or continuous.

SUPPORTING SKILLS

I	Distinguish discrete inputs from values varying continuously over an interval. SK-FUNC-DISCRETE-CONTINUOUS · Introduce
---	---